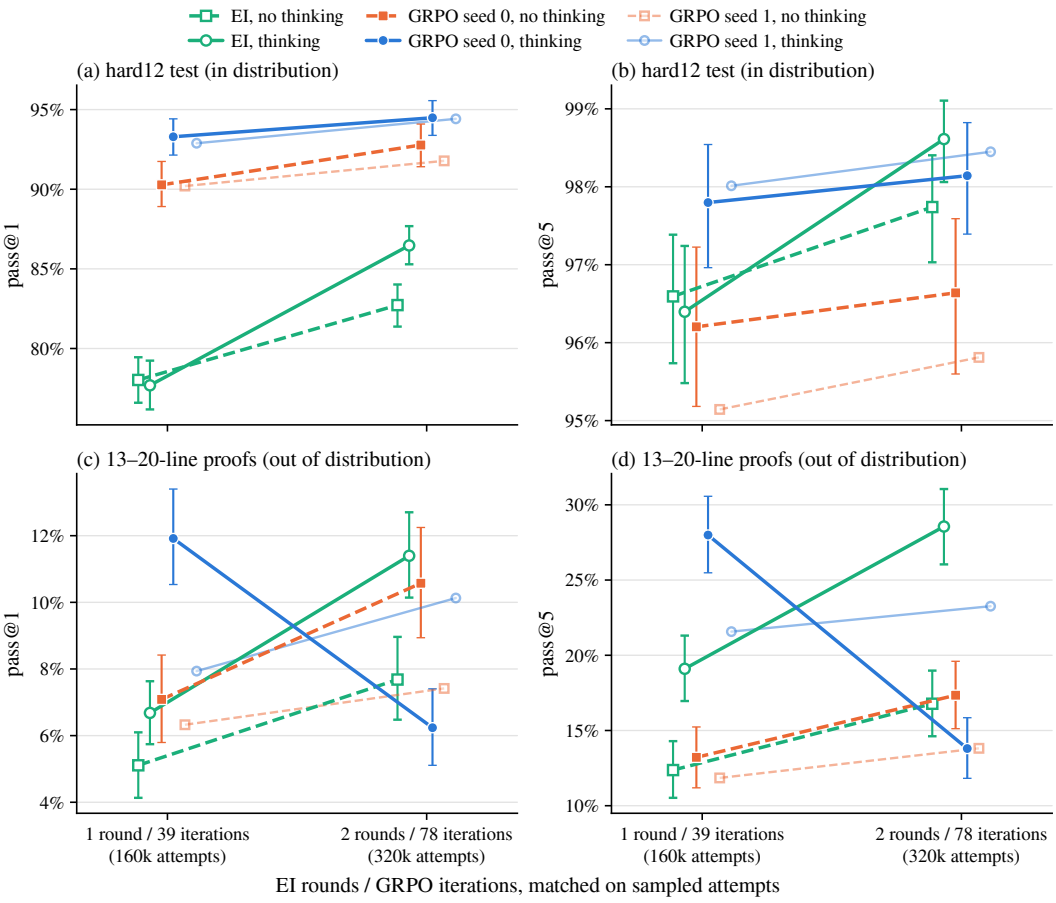
EI, no thinking
EI, thinking
GRPO seed 0, no thinking
GRPO seed 0, thinking
GRPO seed 1, no thinking
GRPO seed 1, thinking
(a) hard12 test (in distribution)
(b) hard12 test (in distribution)
(c) 13–20-line proofs (out of distribution)
(d) 13–20-line proofs (out of distribution)
pass@1
pass@5
95%
90%
85%
80%
99%
98%
97%
96%
95%
12%
10%
8%
6%
4%
30%
25%
20%
15%
10%
1 round / 39 iterations
(160k attempts)
2 rounds / 78 iterations
(320k attempts)
EI rounds / GRPO iterations, matched on sampled attempts